

Problem Set 2

Due on Canvas at 12:00 noon Friday September 27

Instructions: **No partial credit will be awarded on quantitative questions.** All answers must be entered in the Problem Set 1 Quiz on Canvas. Note that it is **not possible to edit your responses** on Canvas after submitting. Best practice is to **complete the problem set on paper** and then transfer solutions to Canvas. You may collaborate on this problem set with other students in your learning team, but each student must submit his or her own solutions.

Grading: Grading is done on a ten point scale across all problems in the problem set. Note that after submitting on canvas, you may see a score, but that does not include any points for any qualitative questions and is only a preliminary score. Grades will be finalized a few days after the due date.

Material covered: This problem set covers the production and perfect competition lectures of the course (Sept 12-Sept 17).

Problem 1. “Cobb-Douglas” Production of Soccer Balls

Soccer balls are produced in Bangladesh by using a mix of capital (K, machines) and labor (L, employees) according to the following production function:

$$Q = 2L^{3/4}K^{1/4}$$

where K and L are measured in hours (note: the “power rule” for derivatives applies even to fractional exponents¹). An hour of labor costs \$20 and an hour of capital costs \$40 (the currency is the Bangladeshi Taka, but we will use the \$ sign). Throughout you can verify your answers with Excel/Solver.

- a. What is the optimal level of labor to employ as a function of the amount of capital?

Answer:

$$\begin{aligned} MP_L &= 1.5L^{-\frac{1}{4}}K^{\frac{1}{4}} \\ MP_K &= 0.5L^{\frac{3}{4}}K^{-\frac{3}{4}} \\ \frac{1.5L^{-\frac{1}{4}}K^{\frac{1}{4}}}{20} &= \frac{0.5L^{\frac{3}{4}}K^{-\frac{3}{4}}}{40} \\ 60L^{-\frac{1}{4}}K^{\frac{1}{4}} &= 10L^{\frac{3}{4}}K^{-\frac{3}{4}} \\ 6K &= L \end{aligned}$$

- b. Assume that the firm wishes to produce 230 balls per hour at minimum cost. Assuming it can freely vary its use of capital and labor, how much capital and labor should it employ?

Answer:

$$\begin{aligned} Q = 230 &= 2(6K)^{3/4}K^{1/4} \\ K &= 30 \\ L &= 180 \end{aligned}$$

- c. What is the total cost of producing the 230 soccer balls each hour? (In Bangladeshi Taka, but again we use the symbol \$)

Answer:

$$Cost = \$4800$$

- d. What is the average cost of producing the 230 units?

Answer:

¹ If you're unsure about your derivative calculation, feel free to verify at sites like <https://www.derivative-calculator.net>, where you would just need to rename “x” the variable you are differentiating with respect to.

$$AC = \$20.87$$

- e. Discuss your optimal production plan, making sense of the labor-to-capital ratio that you have solved for.

Answer:

Note that in the production function, Labor is more effective, and it is also less expensive than Capital. The 6:1 ratio therefore makes sense.

- f. What is the marginal product of labor relative to the price of labor evaluated at the quantity of labor you found in part a? (Use Solver if you want.)

Answer:

Increasing Labor by 1 yields 1 more output, at an increased cost of \$20.

$$\begin{aligned} MP_L &= 1.5(180)^{-\frac{1}{4}}(30)^{\frac{1}{4}} \\ &= 0.958 \end{aligned}$$

- g. What is the marginal product of capital relative to the price of capital evaluated at the quantity of capital you found in part a? (Use Solver if you want.)

Answer:

Increasing Capital by 1 yields 2 more output, at an increased cost of \$40.

$$\begin{aligned} MP_K &= 0.5(180)^{\frac{3}{4}}(30)^{-\frac{3}{4}} \\ &= 1.917 \end{aligned}$$

- h. Now suppose that the level of capital is fixed at the level you identified in part a (i.e. you cannot adjust your capital usage in the short run), but you want to produce 300 units of output instead of 230. How much labor should you employ to do this? What has happened to average cost? What has happened to the marginal products of capital and labor relative to their costs? (Again use Solver as needed.) What is going on here?

Answer:

The solution now uses 256.5 units of labor. Cost is 6330, and average cost is \$21.10. We are no longer at an optimal point as we should be using 6x labor over capital and are not. MPI for labor is higher than MPI for capital.

Problem 2. E-Scooters. The e-scooter craze is sweeping the nation, with cities like Los Angeles, Chicago, and Atlanta overrun with dockless electric scooters that can be rented via an app. A New Haven startup, Scüt, is hoping to bring the craze to the Elm City. They have determined that their daily cost function (where quantity is in number of scooters placed around the city on a given day) is

$$TC(q) = 80 + 20q + 0.05q^2$$

Unfortunately for Scüt, the e-scooter business is perfectly competitive, and they may face rivals such as Lime, Bird, and even Uber and Lyft in the future.

(a) What is Scüt's supply function for scooters (for the range of prices where they are willing to supply any), **as a function of the market price p (which is daily revenue per scooter) for an e-scooter?**

Supply function: $Q_S(P) =$ _____

Answer

Given the cost function, marginal cost is

$$MC(q) = \frac{dTC(q)}{dq} = 20 + 0.1q$$

since it is in a perfectly competitive market, they will be price takers and set price equal to marginal cost

$$P = 20 + 0.1q$$

since this expression relates the number of scooters they will supply and the price they receive, we can solve this for q to get their supply function

$$q = -200 + 10P$$

(b) If the current prevailing market price (daily revenue) in the short run is \$40 per day per scooter, how many scooters will Scüt supply, and what will their daily profit be?

Supplied Quantity (dozens): $Q_S(P) =$ _____

Daily Profit: _____

Answer

Given their supply function, at the prevailing market price of \$40 they will supply

$$Q_S(P) = -200 + 10 \times 40 = 200$$

and their daily profits are

$$\pi = P \times Q - TC(Q) = P \times Q - 80 - 20 \times Q - 0.05Q^2$$

$$\pi = 40 \times 200 - 80 - 20 \times 200 - 0.05(200)^2 = \$1,920$$

(c) Scüt's success has attracted some attention, leading to a price war in New Haven. In the short-run, the fixed cost component of total costs is unavoidable. What is the price below which Scüt would stop placing scooters around the city in the short term?

Price _____

Answer

Given that Scüt has fixed costs that are unavoidable in the short run, they will still produce (even if they make negative profits) as long as

$$\min(AVC) < P$$

The average variable cost curve is

$$AVC = \frac{TVC}{q} = 20 + 0.05q$$

this cost curve is minimized when q is as small as it can be, near zero (technically, the average variable cost curve is not defined at $q = 0$). For simplicity, just ignore this technicality so that $\min(AVC) = 20$. If you make the assumption that they can only produce discrete numbers of scooters, then $q = 1$ is as small as it can be and $\min(AVC) = 20.05$. So if they face a price below \$20 (\$20.05) they will stop supplying entirely in the short run.

(d) In the long run, there is free entry, and indeed several competitors join the market with the same cost function as Scüt. What will be the long-run equilibrium price (daily revenue) per e-scooter?

Price _____

Answer

In the long run, the output decision is the same as in the short run, so the supply function doesn't change. However, since there is free entry and positive profits to be made, firms will enter and exit until the long-run equilibrium price is

$$P = \min(ATC)$$

The average total cost is

$$ATC = \frac{TC(q)}{q} = \frac{80}{q} + 20 + 0.05q$$

to find the value of q where the ATC is minimized, take the derivative of ATC with respect to q and solve the first order condition for q :

$$\frac{\partial ATC}{\partial q} = -\frac{80}{q^2} + 0.05 = 0$$

$$0.05q^2 = 80 \rightarrow q^* = 40$$

so plugging $q^* = 40$ into the ATC function will tell us the minimal value of the ATC :

$$\min(ATC) = \frac{80}{q^*} + 20 + 0.05q^* = 2 + 20 + 2 = 24$$

so $P = \min(ATC) = 24$ is the long-run equilibrium price.

Alternatively, leverage the fact that $\min(ATC)$ occurs when $MC=ATC$,

$$20 + 0.1q = \frac{80}{q} + 20 + 0.05q$$

$$0.05q^2 = 80$$

$$q^2 = 1600$$

$$q = 40$$